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Algorithmic Methods of Data Mining

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1 Introduction to data mining

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1.1 Course organization

The course has two parts. The first part brings everyone to the same level, especially those who have not studied computer science before: basic computer architecture, data structures (different problems call for different ways of storing data), algorithms (how to solve a problem quickly, or even optimally, and how to measure its complexity), and programming, practised in the lab, in the homeworks and in the hackathon. The second part covers data mining topics: text mining, clustering and graphs.

1.2 What is data mining?

Huge amounts of data are produced in every field: social networks, for instance, both contain and generate an enormous quantity of data. These data hide patterns that we would like to find, but the amount is far too large to inspect by hand. Data mining tools let us explore data that we could not otherwise process.

Definition 1.1: Data mining

Data mining is the use of efficient techniques for the analysis of very large collections of data and the extraction of useful and possibly unexpected patterns in data.

We will meet two kinds of tasks:

- **Descriptive tasks** summarize the data and reveal their structure, for example groups of similar objects.

- **Predictive tasks** use the data to predict something unknown, for example whether a person will click on an advertisement.

1.3 Structured, unstructured and semi-structured data

Data come with different amounts of organization.

Definition 1.2: Structured, unstructured and semi-structured data

- *Structured data* have a clean, fixed form. For example, a university's student records store a name, a surname and a student ID for every student. These are the best data we can hope for.
- *Unstructured data* have no predefined form, for example text and pages found on the internet. Most of the data we meet are of this kind.
- *Semi-structured data* lie in between: they have some structure, but it can be modified freely. Spreadsheets (Excel) and XML files are typical examples.

Note: Data cleaning

Data must be cleaned before they are analysed: if the data are dirty, the algorithm will not work. Cleaning used to be very hard to do by hand; today software tools help, but we should still check that the result is correct. Semi-structured data are especially prone to errors, because their structure is easy to break.

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1.4 Types of data

Data are very heterogeneous. Common types include:

- **Table (record) data**, such as spreadsheets; also images and time series.
- **Transaction data**: each record is a transaction between two parties, such as a credit-card payment or a shopping basket.
- **Document data**: Wikipedia pages, tweets, ...
- **Network data**: the web, where pages are linked to one another. Google works with these graphs, and Wikipedia's internal links form one as well.
- **Genomic sequences**: databases of DNA sequences, often used to find correlations with diseases.
- **Climate data**.
- **Behavioural data**: everything recorded about us, for example by smartphones.

1.5 Data as mathematical objects

1.5.1 Records, attributes and sparsity

In table data, each *row* describes an object (for example a person) and each *column* describes an *attribute* of the object.

Definition 1.3: Sparsity

A data table is *sparse* when most of its cells are empty (or zero). The *sparsity* of a table describes how many of its cells are filled, that is, whether many or few values are missing.

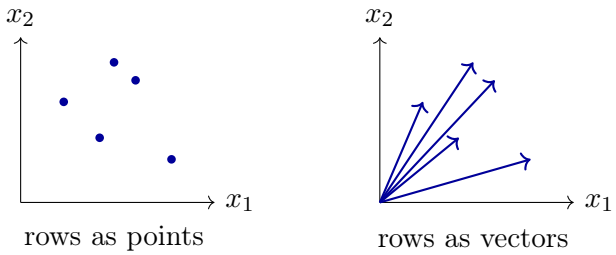
Attributes can be of different types:

- **Binary**: two possible values, such as yes/no.
- **Categorical**: values in a finite set of classes, such as a colour or a school grade like "good" or "bad".
- **Numeric**: numbers, either *discrete* (counts) or *continuous* (time, length, temperature).

We can also move between types: a numeric attribute can be turned into a categorical one by grouping values into classes, and categories can be encoded as numbers.

1.5.2 Numeric records as points and vectors

To analyse data with algorithms, we represent them as mathematical objects. Suppose a table has 2000 rows (people) and 7 numeric attributes. Each row is then an element of $\mathbb{R}^7$, which we can view either as a *point* in a 7-dimensional space or as a *vector* starting from the origin.



1.5.3 Document data

A text contains many words. We represent a document as a vector with one dimension per word of the vocabulary; each entry counts how many times that word appears in the document.

Example 1: Documents as word-count vectors

	word 1	word 2	word 3	⋯
Document 1	10	7	1	⋯
Document 2	15	5	0	⋯

The vocabulary is large, and each document uses only a small part of it. Most entries are therefore 0, so document data have a very high sparsity.

1.5.4 Transaction data

Each transaction has an identifier (TID) and the set of items it involves.

Example 2: A transaction table

TID	Items
1	bread, coke, milk
2	milk, apples
3	carrots
4	chocolate, honey

1.5.5 Ordered data and graph data

In *ordered data* the order of the entries is essential. A genomic sequence such as `GGTTCCGCTTCA`... is an ordered string of characters; time series are another example. In *graph data*, objects are nodes and relations are edges. The classic example is the web graph, whose nodes are pages and whose edges are links.

2

Analysis of algorithms

2.1 Problems and algorithms

A computational *problem* specifies which inputs are allowed and which output each input should produce. An *algorithm* is a finite sequence of steps that produces the required output for every input. Our first problem is finding the maximum of an array.

Problem: Max-Find

Input: an array $A = (A[1], A[2], \ldots, A[n])$ of n numbers.
Output: the maximum value in A.

Note: Indexing

In pseudocode we index arrays from 1, so $A[i]$ is the i-th element. In Python, indexing starts from 0: for $\vec{x} = (x_1, x_2, x_3)$ we have `x[0]` $= x_1$. The length of the array is `n = len(A)`.

Algorithm 2.1: Max-Find

```
1: function MaxFind(A)
2:     max ← A[1]
3:     for i = 1 to n do
4:         if max < A[i] then
5:             max ← A[i]
6:         end if
7:     end for
8:     return max
9: end function
```

The loop may also start from $i = 2$, since comparing $A[1]$ with itself is useless. For every algorithm we always ask two questions:

1. **Correctness:** does it solve my problem?
2. **Efficiency:** how fast is it?

2.2 Correctness

Testing an algorithm on some inputs is not enough: it may work only on the inputs we tried. To show that an algorithm is *wrong*, a single counterexample suffices. To show that it is *correct*, we need a proof, typically by induction on the number of steps performed.

Proposition 2.1: Correctness of Max-Find

After the iteration with index i, the variable max in Algorithm 2.1 equals

$$\max\{A[1], \ldots, A[i]\}.$$

In particular, when the loop ends ($i = n$), the algorithm returns the maximum of A.

Proof. We use induction on i.

Base case ($i = 1$): $max = A[1]$, which is the maximum of $\{A[1]\}$.

Inductive step: assume that after iteration i we have $max = \max\{A[1], \ldots, A[i]\}$. In iteration $i + 1$,

$$A = \big[\underbrace{A[1], A[2], \ldots, A[i]}_{\text{maximum}=max}, A[i+1], \ldots, A[n]\big].$$

If $A[i+1] > max$, then $A[i+1]$ becomes the new maximum; otherwise we keep the previous one. In both cases $max = \max\{A[1], \ldots, A[i+1]\}$. □

2.3 Efficiency: counting steps

How long an algorithm takes depends on many things: the program, but also the input, the programming language, the software, the operating system and the hardware. To compare algorithms, we ignore all of these and look only at the algorithm itself: we count how many elementary steps it performs.

Definition 2.1: Running time

The *running time* $T(n)$ of an algorithm is the number of steps it requires on an input of size n. Unless stated otherwise, we consider the *worst case*: the largest number of steps over all inputs of size n.

Example 3: Counting the steps of Max-Find

```
max ← A[1]                                   ▷ 1 step
for i = 1 to n do                            ▷ n steps
    if max < A[i] then                       ▷ n steps
        max ← A[i]                           ▷ k ≤ n steps
    end if
end for
return max                                   ▷ 1 step
```

Hence $T(n) = 2n + k + 2$. The number k of updates depends on the input:

- *Best case*: the maximum is $A[1]$, so the update never runs ($k = 0$).
- *Worst case*: the array is sorted in increasing order, so the update runs every time ($k = n$).

In the worst case,

$$T(n) = 2n + n + 2 = 3n + 2.$$

The same algorithm can be written with a `while` loop:

Example 4: Max-Find with a while loop

```
max ← A[1]                                   ▷ 1 step
while i ≤ n do                               ▷ n steps
    if A[i] > max then                       ▷ n steps
        max ← A[i]                           ▷ ≤ n steps
    end if
    i ← i + 1                                ▷ n steps
end while
return max                                   ▷ 1 step
```

In the worst case $T(n) = 4n + 2$.

Both versions are correct and do the same thing, and both running times are linear functions of n. The exact constants depend on how we count: initializing i is one more step, and the loop test runs once more when it finally fails. Such details change only the constants, which is why we will learn to ignore them.

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2.4 Asymptotic notation

In the previous section we counted the steps of two versions of Max-Find, one with a `for` loop and one with a `while` loop, and obtained $T(n) = 3n + 2$ and $T(n) = 4n + 2$. Which one is faster? The exact numbers depend on how we count, so comparing them directly

tells us little. What matters is how the running time changes as the input grows: if n doubles, both running times roughly double. In this sense, both *grow like* n.

Asymptotic ("big O") notation makes this idea precise. It ignores constant factors and lower-order terms: the $+2$ can be dropped, and $4n$ grows faster than $3n$ only by a constant factor, which makes no real difference.

Definition 2.2: O, Ω and Θ

Let $T, g : \mathbb{N} \to \mathbb{R}_{\geq 0}$.

- $T(n) = O(g(n))$ if there are constants $c > 0$ and n_0 such that $T(n) \leq c\, g(n)$ for all $n \geq n_0$: g is an *upper bound* on the growth of T.
- $T(n) = \Omega(g(n))$ if there are constants $c > 0$ and n_0 such that $T(n) \geq c\, g(n)$ for all $n \geq n_0$: g is a *lower bound*.
- $T(n) = \Theta(g(n))$ if both $T(n) = O(g(n))$ and $T(n) = \Omega(g(n))$: T grows exactly like g, up to constants.

Informally, O means "at most", Ω means "at least" and Θ means "exactly", always up to constant factors and for large n. The running time of Max-Find is $T(n) = \Theta(n)$: it grows like a linear function of n.

Proposition 2.2: Combining upper and lower bounds

$$T(n) = O(g(n)) \text{ and } T(n) = \Omega(g(n)) \iff T(n) = \Theta(g(n)).$$

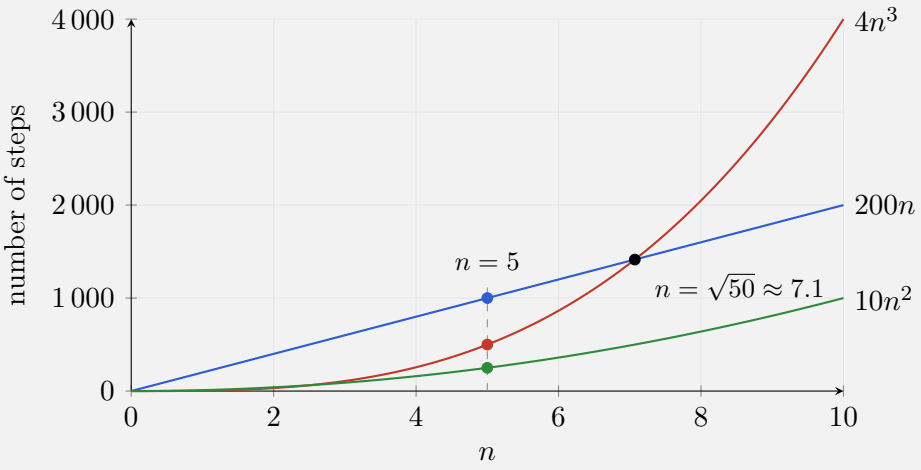
This is the asymptotic version of "$x \leq 3$ and $x \geq 3$ imply $x = 3$". For instance, $T(n) = O(n^3)$ and $T(n) = \Omega(n^3)$ give $T(n) = \Theta(n^3)$.

Upper bounds are not unique: if $T(n) = \Theta(n^2)$, then also $T(n) = O(n^2)$, $T(n) = O(n^3)$, $T(n) = O(n^3 \log n)$, and so on.

Example 5: The dominant term

Let $T(n) = 4n^3 + 10n^2 + 200n + 7$. How fast does this function grow?

The plot compares the three terms that depend on n.



For small n, a lower-order term can be the largest. At $n = 5$ we have

$$200n = 1000, \qquad 4n^3 = 500, \qquad 10n^2 = 250.$$

As n grows, however, the cubic term overtakes $200n$ at $n = \sqrt{50} \approx 7.1$ and dominates from then on, so it determines the running time.

Upper bound. For $n \geq 1$, every power of n in $T(n)$ is at most n^3, so

$$T(n) \leq 4n^3 + 10n^3 + 200n^3 + 7n^3 = 221\, n^3.$$

Hence $T(n) = O(n^3)$, with $c = 221$ and $n_0 = 1$.

Lower bound. All the terms are non-negative, so

$$T(n) \geq 4n^3.$$

Hence $T(n) = \Omega(n^3)$, with $c = 4$ and $n_0 = 1$.

Conclusion. Therefore, by Proposition 2.2, $T(n) = \Theta(n^3)$.

Other upper bounds. For $n \geq 1$,

$$T(n) \leq 221\, n^3 \leq 221\, n^4 \leq 221\, n^5,$$

so we also have $T(n) = O(n^4)$ and $T(n) = O(n^5)$. However, $T(n) \neq \Theta(n^4)$: since $T(n)/n^4 \leq 221/n \to 0$, no constant $c > 0$ gives $T(n) \geq c\, n^4$ for all large n.

When analysing an algorithm we are mostly interested in the *upper bound*, that is, in the worst case. Often we do not know exactly how many times a step runs. In Algorithm 2.1, we do not know in advance how many times the update $max \leftarrow A[i]$ will run, but in the worst case (a sorted array) it runs n times, so it never runs more than n times. Bounding each step from above gives an upper bound on $T(n)$.

2.4.1 Comparing growth rates

Proposition 2.3: Growth rates

For large n (that is, up to constants and eventually):

- $n^c \leq n^d$ for $c \leq d$;
- $\log^c n = (\log n)^c \leq n^d$ for all constants $c, d > 0$;
- $n^c \leq d^n$ for every constant c and every $d > 1$;
- more generally, $n^c \leq d^{an^b}$ for $d > 1$ and constants $a, b > 0$.

Consequently,

$$\begin{aligned} 1 \leq \log\log n \leq \log n \leq \log^2 n \leq \sqrt{n} \leq n \\ \leq n \log\log n \leq n \log n \leq n\sqrt{n} \leq n^2 \leq n^3 \leq 2^n. \end{aligned}$$

For example, $(\log n)^{10}$ and $\sqrt{n}$: the logarithm raised to any power always loses eventually, so $\sqrt{n}$ grows faster.

Note: The base of the logarithm does not matter

Changing the base of a logarithm only multiplies it by a constant:

$$\log_2 n = \frac{\log_{10} n}{\log_{10} 2} = \underbrace{\frac{1}{\log_{10} 2}}_{\text{constant}} \cdot \log_{10} n.$$

Asymptotically all bases are equivalent, so we simply write $O(\log n)$.

An exponential running time means that the algorithm is extremely slow: an algorithm with exponential running time is hopeless except for tiny inputs. Ideally we would like logarithmic or linear running times.

n	$\log_2 n$	$n \log_2 n$	n^2	2^n
10	3.3	33	10^2	$\approx 1.0 \cdot 10^3$
100	6.6	664	10^4	$\approx 1.3 \cdot 10^{30}$
1000	10.0	9966	10^6	$\approx 1.1 \cdot 10^{301}$

2.5 Lower bounds and optimal algorithms

Max-Find is correct and runs in time $\Theta(n)$. Could we do better, say $O(\sqrt{n})$? No: any correct algorithm must read every element, because an element it never reads could be the maximum. With n elements to read, no algorithm can beat $\Omega(n)$.

Definition 2.3: Lower bound and optimal algorithm

A problem has *lower bound* $\Omega(g(n))$ if every algorithm that solves it needs $\Omega(g(n))$ steps in the worst case. An algorithm is (asymptotically) *optimal* if its running time matches a lower bound for the problem.

Example 6: Max-Find is optimal

Max-Find (Algorithm 2.1) runs in $O(n)$, as we computed in Example 3, and the problem requires $\Omega(n)$, so Max-Find is *optimal*.

2.6 Prefix sums

Problem: Prefix sums

Input: an array A of length n.
Output: an array S of length n such that, for every i,

$$S[i] = \sum_{j=1}^{i} A[j].$$

For example, $A = [3, 6, -2, -5, 8]$ gives $S = [3, 9, 7, 2, 10]$.

2.6.1 A quadratic algorithm

The direct approach computes each sum from scratch.

Algorithm 2.2: Prefix sums, direct version

```
1: n ← length(A)                                  ▷ 1 step
2: for i = 1 to n do                              ▷ n steps
3:     S[i] ← 0                                   ▷ n steps
4:     for j = 1 to i do                          ▷ i times for the i-th element
5:         S[i] ← S[i] + A[j]                     ▷ i times
6:     end for
7: end for
8: return S                                       ▷ 1 step
```

The inner loop runs once for the first element, twice for the second, and n times for the last. Each of its two lines therefore runs

$$1 + 2 + 3 + \cdots + n = \sum_{i=1}^{n} i = \frac{n(n+1)}{2}$$

times, and

$$T(n) = 1 + n + n + 2 \cdot \frac{n(n+1)}{2} + 1 = n^2 + 3n + 2 = \Theta(n^2).$$

Note: A quicker upper bound

We do not need the exact sum. For every i, the inner loop runs at most n times, so

$$T(n) \leq 2n + n \cdot (2n) + 2 = O(n^2).$$

Bounding each loop by its largest possible number of iterations is often enough to obtain an upper bound.

2.6.2 Closing the gap

Is this algorithm optimal? No: the lower bound for the problem is $\Omega(n)$ (we must read every element), while our algorithm takes $\Theta(n^2)$. There is a *gap*, and we can try to close it by finding a better algorithm. The key observation is that each prefix sum extends the previous one:

$$S[i] = \sum_{j=1}^{i} A[j] = \sum_{j=1}^{i-1} A[j] + A[i] = S[i-1] + A[i].$$

Algorithm 2.3: Prefix sums, fast version

```
1: n ← length(A)
2: S[1] ← A[1]
3: for i = 2 to n do
4:     S[i] ← S[i − 1] + A[i]                    ▷ n − 1 times
5: end for
6: return S
```

Now $T(n) = \Theta(n)$, which matches the lower bound: the fast version is optimal.

2.7 An open gap: matrix multiplication

Let $A, B \in \mathbb{R}^{n \times n}$ and $C = AB$. Each entry is

$$C[i,j] = \sum_{k=1}^{n} A[i,k]\, B[k,j].$$

Computing one entry takes n multiplications, and there are n^2 entries, so the direct algorithm runs in $\Theta(n^3)$.

A lower bound is $\Omega(n^2)$: the output matrix C has n^2 values, and we must compute each of them. For many years, researchers have tried to close the gap between n^2 and n^3. The gap has been reduced, but it has not yet been closed.

Note: For reference

Strassen's algorithm (1969) multiplies two $n \times n$ matrices in $O(n^{\log_2 7}) \approx O(n^{2.81})$ operations. The best known exponent today is about 2.37, still above the lower bound 2.

3 Recursion

3.1 Fibonacci numbers

Definition 3.1: Fibonacci sequence

$$\text{fib}(n) = \begin{cases} 1, & n = 0 \text{ or } n = 1, \\ \text{fib}(n-1) + \text{fib}(n-2), & n \geq 2. \end{cases}$$

n	0	1	2	3	4	5	6	7	8	9
$\text{fib}(n)$	1	1	2	3	5	8	13	21	34	55

The definition is *recursive*: it defines $\text{fib}(n)$ in terms of smaller values. It translates directly into a recursive algorithm, one that calls itself.

Algorithm 3.1: Recursive Fibonacci

```
1: function FIBONACCI(n)
2:     if n = 0 or n = 1 then                          ▷ 1 step
3:         return 1                                    ▷ at most 1 step
4:     end if
5:     f1 ← FIBONACCI(n − 1)                           ▷ T(n − 1) steps
6:     f2 ← FIBONACCI(n − 2)                           ▷ T(n − 2) steps
7:     return f1 + f2                                  ▷ 1 step
8: end function
```

3.1.1 Running time of the recursive algorithm

Let $T(n)$ be the number of steps for input n. Counting the steps of Algorithm 3.1 gives a *recurrence*:

$$T(n) = \begin{cases} 2, & n = 0 \text{ or } n = 1, \\ T(n-1) + T(n-2) + 3, & n \geq 2. \end{cases}$$

The formula is recursive. How can we solve it? A first idea is to expand it. The recurrence holds for *every* $n \geq 2$, so we can apply it to the terms on the right as well: replacing n with $n-1$ gives $T(n-1) = T(n-2) + T(n-3) + 3$, and replacing n with $n-2$ gives $T(n-2) = T(n-3) + T(n-4) + 3$. In the computations below, each underlined term is replaced, in the next line, by the expression underlined in the same colour.

First expansion.

$$\begin{aligned} T(n) &= \underline{T(n-1)} + \underline{T(n-2)} + 3 \\ &= \underline{\big[T(n-2) + T(n-3) + 3\big]} + \underline{\big[T(n-3) + T(n-4) + 3\big]} + 3 \\ &= T(n-2) + 2T(n-3) + T(n-4) + 9. \end{aligned}$$

Second expansion. We expand each of the three terms again:

$$\begin{aligned} T(n) &= \underline{T(n-2)} + \underline{2T(n-3)} + \underline{T(n-4)} + 9 \\ &= \underline{\big[T(n-3) + T(n-4) + 3\big]} + \underline{2\big[T(n-4) + T(n-5) + 3\big]} \\ &\quad + \underline{\big[T(n-5) + T(n-6) + 3\big]} + 9 \\ &= T(n-3) + 3T(n-4) + 3T(n-5) + T(n-6) + 21 = \cdots \end{aligned}$$

The expansion stops when the arguments reach 0 or 1. After two expansions the smallest argument is already $n-6$, and the number of terms keeps growing: we may suspect that $T(n)$ is exponential.

Proposition 3.1: The recursive algorithm is exponential

The running time of Algorithm 3.1 satisfies

$$T(n) = \Omega\big(2^{n/2}\big) = \Omega\big((\sqrt{2})^n\big).$$

Proof. T is non-decreasing: $T(1) = T(0)$, and for $n \geq 2$ we have $T(n) \geq T(n-1)$.

Replacing $T(n-1)$ with the smaller $T(n-2)$,

$$T(n) = T(n-1) + T(n-2) + 3 \geq 2\,T(n-2) + 3.$$

Applying the same inequality repeatedly,

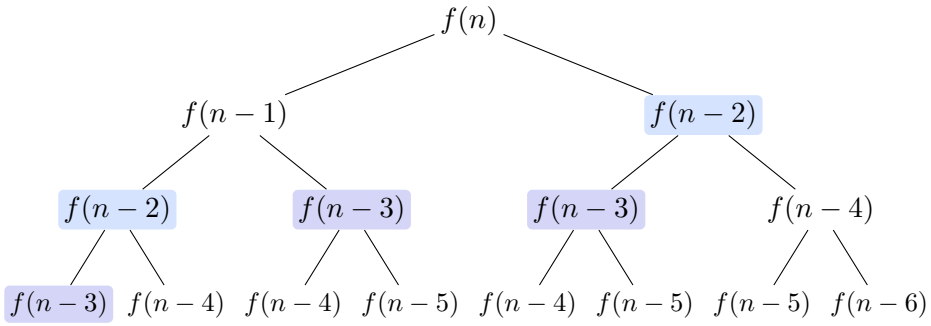
$$\begin{aligned}
T(n) &\geq 2\,\big(2T(n-4)+3\big) + 3 = 4\,T(n-4) + 9 \\
&\geq 4\,\big(2T(n-6)+3\big) + 9 = 8\,T(n-6) + 21 \\
&\geq 8\,\big(2T(n-8)+3\big) + 21 = 16T(n-8) + 45 \\
&\;\;\vdots \\
&\geq 2^i T(n-2i) + 3(2^i - 1) \geq 2^i T(n-2i) \qquad \text{after } i \text{ steps.}
\end{aligned}$$

For even n, take $i = n/2$:

$$T(n) \geq 2^{n/2}\,T(0) = 2 \cdot 2^{n/2} = 2(\sqrt{2})^n.$$

For odd n, take $i = (n-1)/2$ and use $T(1) = 2$ in the same way. □

Why is it so slow? To compute fib(n), the algorithm makes two calls, fib$(n-1)$ and fib$(n-2)$, each of which makes two more calls, and so on. It computes the same values a lot of times, again and again: the work roughly doubles at each level of the recursion tree.



Here $f(k)$ stands for a call to fib(k). Already in the first levels, $f(n-2)$ is computed twice and $f(n-3)$ three times.

3.1.2 Avoiding repeated work

There are two ways to fix the problem:

- **Store the values** (memoization): using some memory, we compute each fib(k) once and look it up afterwards.
- **Compute bottom-up**: start from the base cases and fill the table from left to right, $1+1=2$, $2+1=3$, $3+2=5$, $5+3=8$, …, each value being the sum of the previous two.

Algorithm 3.2: Iterative Fibonacci

```
1: function FibonacciIterative(n)
2:     if n ≤ 1 then
3:         return 1
4:     end if
5:     previous ← 1                                   ▷ fib(0)
6:     current ← 1                                    ▷ fib(1)
7:     for i = 2 to n do
8:         new ← previous + current
9:         previous ← current
10:        current ← new
11:    end for
12:    return current
13: end function
```

The loop runs $n-1$ times with a constant amount of work per iteration, so the running time is $T(n) = O(n)$: this algorithm is linear.

Note: Indexing and closed form

The slides use the convention $F_0 = 0$, $F_1 = 1$ (their iterative algorithm returns n for $n \leq 1$ and starts from 0 and 1). With that convention $\text{fib}(n) = F_{n+1}$, so the two sequences differ only by a shift. There is also a closed formula, which we do not prove:

$$F_n = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right], \qquad \text{fib}(n) = F_{n+1}.$$

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3.2 Binary search

Problem: Search

Input: an array A of n numbers and a value x.
Output: TRUE if x is in A, FALSE otherwise.

We could compare x with every element, with linear complexity. If A is *sorted*, we do not need to check all the elements: we look at the central element and compare it with x. If x is larger, it can only be in the right half; if smaller, only in the left half. We then repeat on that half.

Example 7: Searching a sorted array

Let $A = [1, 3, 7, 10, 13, 24, 30, 33, 39, 55]$ and $x = 39$.

1. The central element is $A[5] = 13 < 39$, so we keep the right part $[24, 30, 33, 39, 55]$.
2. Its central element is $33 < 39$, so we keep $[39, 55]$.
3. Its central element is $39 = x$: found.

Algorithm 3.3: Binary search

```
1: function BinarySearch(A, x)
2:     n ← len(A)
3:     if n = 0 then
4:         return FALSE
5:     end if
6:     if x < A[1] or x > A[n] then                ▷ x outside the range of A
7:         return FALSE
8:     end if
9:     mid ← ⌈n/2⌉
10:    if x = A[mid] then
11:        return TRUE
12:    else if x < A[mid] then
13:        return BinarySearch(A[1, ..., mid − 1], x)
14:    else
15:        return BinarySearch(A[mid + 1, ..., n], x)
16:    end if
17: end function
```

The *ceiling* $\lceil y \rceil$ is the smallest integer greater than or equal to y: for example, $\lceil 3 \rceil = 3$ and $\lceil 3.1 \rceil = 4$.

Example 8: An unsuccessful search

Let $A = [13, 17]$ and $x = 15$.

1. Since $13 < 15 < 17$, x lies within the range of A.
2. The central element is $A[1] = 13 < 15$ (here $mid = \lceil 2/2 \rceil = 1$), so we keep the right part $[17]$.
3. Now $15 < 17$, so x is outside the range of $[17]$: the algorithm returns FALSE.

3.2.1 Running time

Every line of Algorithm 3.3 takes constant time, except the recursive call. The call works on an array of half the size, so it takes $T(n/2)$ steps instead of $T(n)$:

$$T(n) = \begin{cases} c_1, & n \leq 1, \\ T(n/2) + c_2, & \text{otherwise}, \end{cases}$$

where c_1 and c_2 are constants. For $n \geq 2$,

$$T(n) = T\left(\tfrac{n}{2}\right) + c_2 = T\left(\tfrac{n}{4}\right) + 2c_2 = T\left(\tfrac{n}{8}\right) + 3c_2 = \cdots = T\left(\tfrac{n}{2^i}\right) + i\,c_2.$$

We stop when the array has size 1, that is, when $n/2^i = 1$, or $i = \log_2 n$. Then $T(n/2^i) = T(1) = c_1$ and

$$T(n) = c_1 + c_2 \log_2 n = \Theta(\log n).$$

Note: Subarrays

We treat $A[1, \ldots, mid - 1]$ as a view of the original array, described by its first and last index. Copying the subarray at every call would cost linear time and destroy the logarithmic bound.

We cannot do better than $\Theta(\log n)$, but we are not able to prove this yet. Splitting into three parts instead of two does not help asymptotically: the recurrence becomes $T(n) = T(n/3) + c_2'$, so $T(n) = O(\log_3 n)$, and $O(\log_3 n) = O(\log_2 n)$ because $\log_3 n = \log_2 n / \log_2 3$.

4

Complexity classes

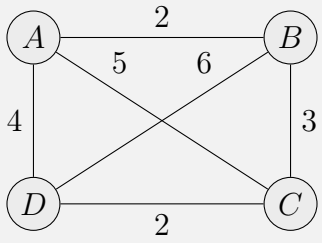
4.1 The traveling salesman problem

We want to visit all the nodes of a weighted graph, starting from a node and coming back to it, while minimizing the total cost. Every order of visit gives a tour, and different tours have different costs.

Problem: Traveling salesman problem (TSP)

Input: n cities and the cost of travelling between each pair of them.
Output: a tour that visits every city exactly once and returns to the starting city, with minimum total cost.

Example 9: Four cities



Starting from A, there are $3! = 6$ orders of visit, which give three distinct tours (each one can be travelled in two directions):

$$A\,B\,C\,D\,A: \ 2+3+2+4=11,$$
$$A\,B\,D\,C\,A: \ 2+6+2+5=15,$$
$$A\,C\,B\,D\,A: \ 5+3+6+4=18.$$

The optimal tour costs 11. In the lecture graph with eight cities, two tours cost 15 and 21.

The brute-force approach tries all tours: with n cities there are $(n-1)!$ orders of visit, and

$$n! = 1 \cdot 2 \cdot 3 \cdots (n-1) \cdot n \geq 1 \cdot 2 \cdot 2 \cdots 2 = 2^{n-1},$$

so the factorial grows even faster than an exponential: this is not good at all. No known algorithm finds the shortest tour quickly.

4.2 Decision problems, P and NP

We usually study *decision problems*, whose answer is yes or no. An optimization problem such as minimizing costs or maximizing profits becomes a decision problem by fixing a threshold. For TSP: *is there a tour that visits all the nodes with total cost at most* 14*?* If the answer is yes, we can ask whether we can do better, for example cost at most 13.

Definition 4.1: The class P

P is the class of problems that we can solve in *polynomial time* (poly-time), that is, in time $O(n^k)$ for some constant k.

Definition 4.2: The class NP

NP is the class of problems for which, if someone gives us a candidate solution, we can verify in polynomial time whether the answer is YES.

Example 10: Easy to verify

- *Rubik's cube*: it is easy to check whether a cube is solved.
- *TSP*: given a tour, we add up its costs and check whether the total is at most 14.
- *Sudoku*: it is easy to check whether a filled grid is correct.

What these problems have in common is that, given a solution, we can verify in poly-time whether it is correct, even though finding one may be hard.

Max-Find and binary search are in P. Every problem in P is also in NP: if we can solve a problem quickly, we can verify an answer by solving it ourselves. Hence $\mathrm{P} \subseteq \mathrm{NP}$.

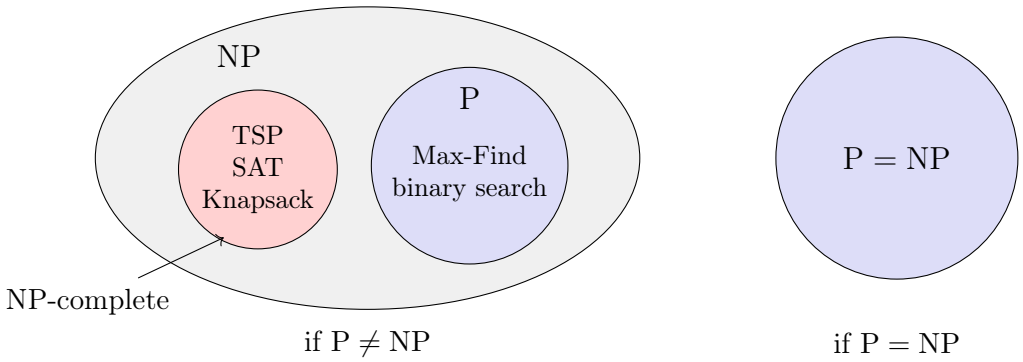
Note: What NP stands for

NP stands for *nondeterministic polynomial time*, not "non polynomial". A problem in NP may well be easy: every problem in P is in NP.

Are there problems in NP that are not in P? Nobody knows.

$$\textit{Is } \mathrm{P} = \mathrm{NP}\textit{, or is } \mathrm{P} \subsetneq \mathrm{NP}\textit{? We don't know.}$$

This is one of the most important open questions in computer science.



4.3 Reductions and NP-completeness

Some problems in NP are as hard as any other problem in NP.

Definition 4.3: NP-complete problem

A problem X is *NP-complete* if:

1. $X \in$ NP;
2. every problem $Y \in$ NP can be *reduced* in polynomial time to X.

A reduction of Y to X solves Y using an algorithm for X plus a polynomial amount of extra work.

Example 11: Max-Find and sorting

If we can sort an array in poly-time, we can also find its maximum in poly-time: sort it and take the last element. Max-Find reduces to sorting.

The converse also holds: if we can find the maximum, we put it in the last position and repeat on the remaining elements, until the whole array is sorted. This uses n calls to Max-Find, a polynomial amount of work, so sorting reduces to Max-Find.

Surprisingly, NP-complete problems exist, such as *SAT* (Boolean satisfiability), which we will not study in detail. If P = NP, the classes collapse: NP-complete problems would be solvable in polynomial time, like every other problem in NP.

Note: Are TSP, Sudoku and Rubik's cube NP-complete?

Complexity classes describe how the cost grows with the input size, so we must consider generalized versions of these puzzles:

- the decision version of TSP ("is there a tour of cost at most k?") is NP-complete;

- Sudoku on $n^2 \times n^2$ grids ("can this partially filled grid be completed?") is NP-complete;
- for the $n \times n \times n$ Rubik's cube, deciding whether a position can be solved in at most k moves is NP-complete (proved in 2018).

The classic 9×9 Sudoku and $3 \times 3 \times 3$ cube have a fixed size, so asymptotic statements do not apply to them directly.

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Corollary 4.1: Solving one NP-complete problem

Let X be NP-complete. If X can be solved in poly-time, then every problem in NP can be solved in poly-time. Equivalently, if some $Y \in \text{NP}$ cannot be solved in poly-time, then neither can X.

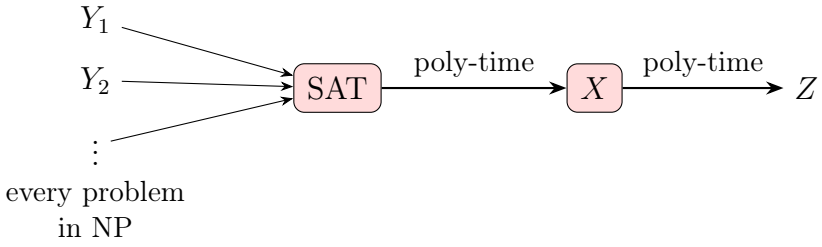
SAT was the first problem proved to be NP-complete: every problem in NP can be reduced to SAT (the Cook–Levin theorem). To show that a new problem is NP-complete, we no longer need to reduce every problem in NP to it.

Proposition 4.1: Proving NP-completeness

To prove that a problem X is NP-complete, show that:

1. $X \in \text{NP}$;
2. there is a problem Y already known to be NP-complete and a poly-time reduction of Y to X.

This works because reductions compose: every problem in NP reduces to Y, and Y reduces to X, so every problem in NP reduces to X. The composition of two polynomials is still a polynomial.



4.4 Some NP-complete problems

4.4.1 Knapsack

We want to fill a backpack with the most valuable set of items without exceeding its capacity.

Problem: Knapsack

Input: a set of items $I = \{1, 2, \ldots, n\}$, where item i has a weight w_i and a value v_i, and a budget B.
Output: a subset of items $S \subseteq I$ such that

$$\sum_{i \in S} w_i \leq B \qquad \text{that maximizes} \qquad \sum_{i \in S} v_i.$$

Example 12: A knapsack instance

Let $B = 55$ and

i	1	2	3	4	5	6	7
w_i	10	20	6	30	15	20	50
v_i	10	15	8	17	12	17	20

We must maximize $\sum_{i \in S} v_i$ subject to the constraint $\sum_{i \in S} w_i \leq 55$. One feasible solution is $S = \{1, 2, 3, 5\}$, with total weight 51 and total value 45. A solution taken "at random" like this one is probably not the best: in Example 16 we will find one of value 47.

4.4.2 Set cover

Suppose we want to rob a casino: we need people who can do different things.

Problem: Set cover

Input: a task that requires the skills $s_1, \ldots, s_n$, and m experts $E_1, \ldots, E_m$, where each $E_i \subseteq \{1, \ldots, n\}$ is the subset of skills that expert i has.
Output: the smallest number of experts that together cover all the skills.

In general, the skills are elements $e_1, \ldots, e_n$ and the experts are sets $S_1, \ldots, S_m$ of elements. We want the minimum number of sets whose union contains all the elements.

Example 13: Covering six skills

	s_1	s_2	s_3	s_4	s_5	s_6
E_1	×		×	×		
E_2	×	×				
E_3		×			×	
E_4	×			×		×
E_5			×			×

We must take E_3, because it is the only expert with skill s_5. E_3 covers s_2 and s_5; no single other expert covers all of s_1, s_3, s_4, s_6, so we need at least 3 experts. One optimal solution is $\{E_3, E_4, E_5\}$.

A heuristic method: take E_3, then repeatedly take the expert who covers the largest number of skills not yet covered. Such solutions are not optimal in general.

4.4.3 Subset sum

Problem: Subset sum

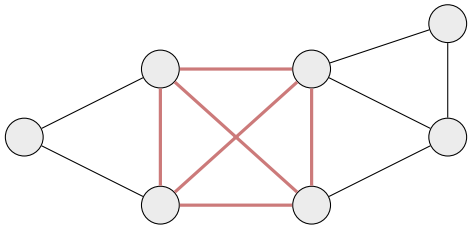
Input: a list of integers.
Output: a nonempty sublist whose elements sum to 0, if one exists.

For the list $[3, 7, -3, -8, 10, -16, 2, 9]$, the sublists $[3, -3]$ and $[7, 9, -16]$ both sum to 0.

4.4.4 Maximum clique

Problem: Maximum clique

Input: a graph.
Output: a largest *clique*, that is, a largest set of nodes that are all connected to one another.



The thick edges form a clique of size 4, the largest in this graph.

Proposition 4.2: Some NP-complete problems

The decision versions of Knapsack, Set cover, Subset sum and Max clique are NP-complete.

We do not prove this: each proof is a poly-time reduction from a known NP-complete problem (Proposition 4.1).

4.5 NP-hard problems

Definition 4.4: NP-hard problem

A problem X is *NP-hard* if every problem in NP can be reduced to X in polynomial time. Unlike an NP-complete problem, X does not have to belong to NP.

NP-hard problems are more general: NP-complete means NP-hard and in NP. For example, the optimization versions of Knapsack and TSP (find the best solution, not just answer yes or no) are NP-hard.

Note: Minimum spanning tree

The minimum spanning tree problem is *not* NP-hard (unless P = NP): Kruskal's and Prim's algorithms solve it in polynomial time. Its relative, the Steiner tree problem, which must connect only a given subset of the nodes, is NP-hard.

5 Approximation and heuristics

For NP-complete and NP-hard problems we do not know fast exact algorithms. Two ways out are approximation algorithms, which come with a guarantee, and heuristics, which do not.

5.1 Approximation algorithms

Definition 5.1: c-approximation algorithm

Let $c \geq 1$. For an input I, let OPT_I be the value of an optimal solution. A *c-approximation algorithm* returns, for every input I, a solution whose value is

- at most $c \cdot \mathrm{OPT}_I$, for a *minimization* problem;
- at least OPT_I / c, for a *maximization* problem.

Example 14: Reading the guarantee

- *TSP (minimization)*: suppose that for an input I the optimal tour costs $\mathrm{OPT}_I = 5000$. A 2-approximation algorithm guarantees a tour of cost at most 10 000: it may cost more than 5000, but we are sure that it does not exceed 10 000. For $c = 1.1$, the tour costs at most $1.1 \cdot 5000 = 5500$, no more than 10% above the optimum.
- *Knapsack (maximization)*: if $\mathrm{OPT}_I = 1000$, a 1.1-approximation guarantees a value of at least $1000/1.1 \approx 909$.

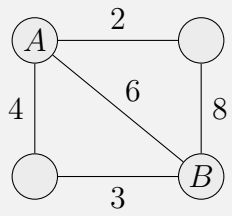
5.1.1 Guarantees without knowing the optimum

How can we prove that a solution is a c-approximation if we cannot find the optimal solution? We may not find OPT_I, but we can often find a *lower bound* $L \leq \mathrm{OPT}_I$ (for a

minimization problem). Then

$$\frac{\text{output}}{\text{OPT}_I} \leq \frac{\text{output}}{L}.$$

Example 15: Using a lower bound



Suppose every solution must travel from A to B. Every route from A to B costs at least 6, so $\text{OPT}_I \geq 6$: 6 is a lower bound. If our algorithm outputs a solution of cost 17, is it good?

$$\frac{\text{output}}{\text{OPT}_I} = \frac{17}{\text{OPT}_I} \leq \frac{17}{6} \approx 2.83,$$

so our solution is at most 17/6 times the optimum.

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5.1.2 How well can we approximate?

Different problems behave very differently:

- For *set cover*, where we minimize the number of sets, a simple greedy algorithm is, at worst, an $O(\log n)$-approximation.
- Many other problems cannot be approximated well in poly-time. For *max clique*, no polynomial-time algorithm can guarantee even a factor like $\sqrt{n}$, unless P = NP.

Some approximation algorithms are also very inefficient. This is why there is another approach: heuristics.

5.2 Heuristics

A *heuristic* is a method that usually finds good solutions quickly, but without any guarantee on their quality.

5.2.1 A greedy heuristic for knapsack

For each item, consider the value per unit of weight

$$r_i = \frac{v_i}{w_i}.$$

We start putting into the backpack the items with the largest r_i, and keep adding items as long as the constraint is satisfied. This usually gives a good solution, but not always the optimal one.

Algorithm 5.1: Greedy knapsack

sort the items so that $r_{i_1} \geq r_{i_2} \geq \cdots \geq r_{i_n}$
$S \leftarrow \varnothing$, $W \leftarrow 0$
for $k = 1$ **to** n **do**
    if $W + w_{i_k} \leq B$ **then**
        $S \leftarrow S \cup \{i_k\}$, $W \leftarrow W + w_{i_k}$
    end if
end for
return S

Example 16: Greedy on the knapsack instance

For the instance in Example 12,

$$r_1 = 1, \quad r_2 = \tfrac{3}{4}, \quad r_3 = \tfrac{4}{3}, \quad r_4 = \tfrac{17}{30}, \quad r_5 = \tfrac{4}{5}, \quad r_6 = \tfrac{17}{20}, \quad r_7 = \tfrac{2}{5}.$$

Item 3 has the largest ratio, so it is the first one we add. The order is $3, 1, 6, 5, 2, 4, 7$:

- add 3 (weight 6), 1 (weight 16), 6 (weight 36) and 5 (weight 51);
- items 2, 4 and 7 no longer fit ($51 + 20 > 55$).

The result is $S = \{1, 3, 5, 6\}$, with weight 51 and value 47. Trying all subsets shows that 47 is optimal for this instance.

The greedy solution is not optimal in general.

Example 17: When greedy fails

Let $B = 100$, ten items with $w_1 = \cdots = w_{10} = 50$ and $v_1 = \cdots = v_{10} = 50$ (each with ratio 1), and an item 0 with $w_0 = 51$ and $v_0 = 52$, whose ratio is slightly larger than 1. Greedy takes item 0 first, and then no other item fits ($51 + 50 > 100$): the value is 52. Taking two items of weight 50 gives value 100.

No known algorithm solves knapsack in poly-time: it is NP-complete. If we could "break" the items and take a fraction of one of them (the *fractional* knapsack), the greedy algorithm would be perfect: we fill the remaining capacity with a fraction of the next item.

5.2.2 Greedy algorithms and meta-heuristics

Definition 5.2: Greedy algorithm

A *greedy* algorithm makes, at each step, the choice that is best for now, hoping that it will also work well in the future, and continues in the same way.

The knapsack heuristic above and the set cover heuristic are both greedy algorithms. Greediness is an example of a *meta-heuristic*: a general strategy, a wider class that includes many heuristics. Another example is *local search*: starting from a solution, such as a tour, we try small modifications (for example, exchanging two of its edges for two others) and keep those that improve it, searching for the optimal one.

6

Distance measures

6.1 Why distances?

Many problems ask us to find distances or similarities between objects. For example, given some documents and a query, we want the documents most similar to the query; or we want groups of similar documents, a *cluster*. The input is a set of objects and the output a group of them. To do this we use different kinds of distances: a function that measures the distance between two elements defines a relation between them.

6.2 Distance functions

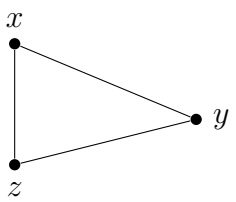
Definition 6.1: Distance function

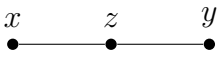
Let U be a universe of elements (documents, objects, strings, ...). A *distance function* is a map

$$d : U \times U \longrightarrow \mathbb{R},$$

taking two elements and returning a number, which satisfies, for all $x, y, z \in U$:

1. $d(x, y) \geq 0$, and $d(x, y) = 0$ if and only if $x = y$;
2. $d(x, y) = d(y, x)$ (symmetry);
3. $d(x, y) \leq d(x, z) + d(z, y)$ (triangle inequality).





equality when z lies between

In the next lectures we will see different distances depending on the data. We should define distances that make sense for the problem: what does it mean for two objects to be "close"? It depends on the problem we are considering.

6.3 L_p distances

Example 18: Restaurants near me

We are designing an app like Google Maps to find restaurants near us. A good distance is the Euclidean one. For $U = \mathbb{R}^m$, $\vec{x} = [x_1, \ldots, x_m]$ and $\vec{y} = [y_1, \ldots, y_m]$,

$$d(\vec{x}, \vec{y}) = \sqrt{\sum_{j=1}^{m} (x_j - y_j)^2}.$$

This is the *Euclidean distance*, or L_2 distance. It is clearly non-negative and symmetric; it also satisfies the triangle inequality, but we do not prove it.

Definition 6.2: L_p distance

For a constant $p \geq 1$,

$$d_p(\vec{x}, \vec{y}) = \left(\sum_{j=1}^{m} |x_j - y_j|^p \right)^{1/p} = \|\vec{x} - \vec{y}\|_p.$$

Changing p gives different distances, which weigh the differences between coordinates differently:

- $p = 2$: the Euclidean distance.
- $p = 1$: the *Manhattan distance*, $\sum_j |x_j - y_j|$.
- $p \to \infty$: the L_∞ distance, $d_\infty(\vec{x}, \vec{y}) = \max_{i \in \{1,\ldots,m\}} |x_i - y_i|$.
- "L_0": the number of coordinates in which the vectors differ, $|\{i : x_i \neq y_i\}|$. For binary strings this is the *Hamming distance*: 00110 and 10100 differ in 2 positions, and $[3, 1, 3]$ and $[0, 1, 3]$ differ in 1.

Example 19: Comparing L_p distances

Let

$$\vec{x} = [3, 7, 0, 2, 1], \qquad \vec{y} = [4, 5, 10, 7, 3], \qquad \vec{z} = [5, 2, 5, 6, 5].$$

The coordinate differences are $|\vec{x} - \vec{y}| = [1, 2, 10, 5, 2]$ and $|\vec{x} - \vec{z}| = [2, 5, 5, 4, 4]$.

p	$\|\vec{x}-\vec{y}\|_p$	$\|\vec{x}-\vec{z}\|_p$
1	$1+2+10+5+2=20$	$2+5+5+4+4=20$
2	$(1+4+100+25+4)^{1/2}\approx 11.58$	$(4+25+25+16+16)^{1/2}\approx 9.27$
3	$(1+8+1000+125+8)^{1/3}\approx 10.45$	$(8+125+125+64+64)^{1/3}\approx 7.28$
∞	10	5

With $p=1$ the two distances are equal. As p increases, both distances decrease, and the larger differences weigh more: $\vec{y}$ differs from $\vec{x}$ by 10 in one coordinate, so it ends up farther than $\vec{z}$, whose differences are moderate everywhere.

6.4 Distances between sets

So far we have worked with vectors. Now imagine that we want to compare the customers of a supermarket based on what they bought.

Example 20: Shopping baskets

Each letter is a product: the first person bought apples (A), bananas (B), carrots (C) and donuts (D):

$$P_1=\{A,B,C,D\},\qquad P_2=\{A,E,G,H\},\qquad P_3=\{B,C,E,F\}.$$

Intuitively, persons 1 and 3 are more similar than persons 1 and 2.

6.4.1 Counting common items

A first idea is to look at the elements in common:

$$|P_1\cap P_2|=1,\qquad |P_1\cap P_3|=2.$$

Is this a good distance? We should check the three properties. Symmetry certainly holds, but the larger the value, the *closer* the two sets: this is a similarity, not a distance.

Note: Distance and similarity

It is often more natural to talk about *similarity* than about distance. The two notions are closely related, but they go in opposite directions: a large similarity means a small distance. Here we do not see a "natural" distance function, and we cannot simply take the negative of the similarity, because distances cannot be negative.

6.4.2 Sets as binary vectors

A second idea is to write each set as a binary vector, with one coordinate per product, and use the L_1 distance:

	A	B	C	D	E	F	G	H
P_1	1	1	1	1	0	0	0	0
P_2	1	0	0	0	1	0	1	1
P_3	0	1	1	0	1	1	0	0

Then $|P_1 - P_2| = [0, 1, 1, 1, 1, 0, 1, 1]$ and $|P_1 - P_3| = [1, 0, 0, 1, 1, 1, 0, 0]$, so

$$\|P_1 - P_2\|_1 = 6, \qquad \|P_1 - P_3\|_1 = 4.$$

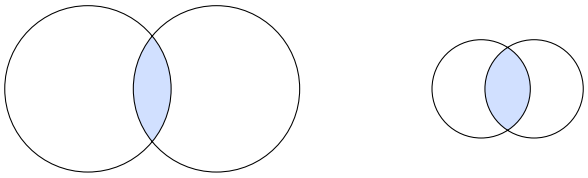
This is how we can turn the comparison into a distance. In general,

$$\|P - Q\|_1 = |P| + |Q| - 2|P \cap Q| = |P \triangle Q|,$$

the size of the *symmetric difference* $P \triangle Q$, the elements in exactly one of the two sets.

6.4.3 Normalizing the intersection

Back to the first idea: the size of the intersection alone is misleading. Two pairs of sets can have intersections of the same size, but if the sets in the second pair are smaller, the intersection is a larger fraction of them, so they are more similar.



1 2

Same intersection, different set sizes: the sets in pair 2 are more similar, because their intersection is a larger fraction of them.

We can normalize the intersection in several ways:

$$\frac{|P \cap Q|}{\max\{|P|, |Q|\}}, \qquad \frac{|P \cap Q|}{|P \cup Q|}, \qquad \frac{|P \cap Q|}{|P| + |Q|}.$$

Definition 6.3: Jaccard similarity

The *Jaccard similarity* of two finite sets P and Q is

$$J(P, Q) = \frac{|P \cap Q|}{|P \cup Q|},$$

where $|\cdot|$ is the cardinality (not the absolute value).

The Jaccard similarity is symmetric and lies between 0 and 1. For the baskets in Example 20,

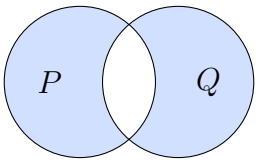
$$J(P_1, P_2) = \frac{|\{A\}|}{|\{A, B, C, D, E, G, H\}|} = \frac{1}{7}, \qquad J(P_1, P_3) = \frac{|\{B, C\}|}{|\{A, B, C, D, E, F\}|} = \frac{2}{6} = \frac{1}{3},$$

so P_1 is more similar to P_3, as expected.

Definition 6.4: Jaccard distance

We turn the Jaccard similarity into a distance by taking

$$d_J(P,Q) = 1 - \frac{|P \cap Q|}{|P \cup Q|} = \frac{|P \cup Q| - |P \cap Q|}{|P \cup Q|} = \frac{|P \triangle Q|}{|P \cup Q|} = \frac{|P \setminus Q| + |Q \setminus P|}{|P \cup Q|}.$$



The symmetric difference $P \triangle Q$: everything except the intersection.

One can prove that the Jaccard distance d_J is a distance in the sense of Definition 6.1. For the baskets, $d_J(P_1, P_2) = 6/7$ and $d_J(P_1, P_3) = 4/6 = 2/3$.

7

Distances between strings

Start of lecture 12 October

7.1 Why compare strings?

Biological data are sequences of bases such as `CTAGGATCGGAC`, and DNA sequences have particular properties: we may want to know how "similar" two parts of DNA are. Search engines need similarity between strings too: if we type "Sceince" instead of "Science" on Google, it still understands what we mean.

Example 21: Which strings are closer?

$X = \texttt{CTAGGAT}, \qquad Y = \texttt{CCAGAGT}, \qquad Z = \texttt{AGCCCTA}.$

Which pair is more similar, X and Y, or X and Z? We would say X and Y. The answer is intuitive, but which property are we using? How do we define a "distance measure" between strings?

7.2 First attempts

- **Compare letter by letter**, as with the Hamming or L_1 distance. But what if the strings have different lengths?
- **Count how many times each character appears.** Biologically, `AAGGGTT` and `TGAAGTG` are very different, even though they contain exactly the same characters.
- **Look at common pieces**: `CATGAGG` and `CAGAGT` share `CA` and `GAG`, even though they have different lengths.

- **Count modifications**: take the second string and count how many changes are needed to make it equal to the first. For `SAPIENZA` and `SAAPIEMZA`, we delete the extra `A` and replace `M` with `N`: two changes, so the distance is 2.

Example 22: Letters as numbers

Associate each letter with its position in the alphabet ($A = 1$, $C = 3$, $G = 7$, $T = 20$) and subtract:

`CAGT` :	3	1	7	20
`ACTT` :	1	3	20	20
difference :	2	2	13	0

$$\Longrightarrow \qquad 2 + 2 + 13 + 0 = 17.$$

The larger the number, the farther the strings; this distance is symmetric. However:

- comparing `SAPIENZA` with `SSAPIENZA`, typed with an extra `S` by mistake, shifts every letter: the two words are almost identical, but the distance explodes;
- the distance between A and T is larger than between A and B only because of the alphabetical order.

This is why such a distance cannot work well.

We go back to the two previous ideas: common pieces (subsequences) and modifications (edit operations). Consider sequences over an alphabet.

7.3 Subsequences

Definition 7.1: Subsequence

A *subsequence* of a sequence S is a sequence that contains a subset of the characters of S in the same order, not necessarily consecutive.

Example 23: Subsequences

For $S =$ `ATCGGACTGA`, all of the following are subsequences: `ATCGGACTGA` itself, `ACGGCTA`, `AAA`, and the empty string.

Definition 7.2: Common subsequence

Given two sequences S and T, a *common subsequence* is a sequence that is a subsequence of both S and T.

For $S =$ `ATGCATG` and $T =$ `AGACTAG`, `AAT` and `AGAG` are common subsequences, and so is the empty string.

7.3.1 The longest common subsequence

Definition 7.3: Longest common subsequence (LCS)

The *longest common subsequence* of S and T is a common subsequence of maximum length. We write $\mathrm{LCS}(S, T)$ for its length.

We can define the similarity of two strings as the length of their LCS.

Example 24: Longest common subsequences

- For `ATGA` and `TAG`, both `AG` and `TG` are longest common subsequences: the LCS need not be unique, and here none is longer than 2.
- Compare `SAPEMZA` with `SAPIENZA` and with `SALERNO`:

  $\mathrm{LCS}(\texttt{SAPIENZA}, \texttt{SAPEMZA}) = 6$ (`SAPEZA`), $\mathrm{LCS}(\texttt{SAPEMZA}, \texttt{SALERNO}) = 3$ (`SAE`).

  `SAPEMZA` has more in common with `SAPIENZA` than with `SALERNO`.
- In Example 21, $\mathrm{LCS}(X, Y) = 5$ (for example `CAGGT`) while $\mathrm{LCS}(X, Z) = 3$ (`CTA`), which confirms our intuition.
- For `CAGTGACCGATAC` and `GACCAGTCAGCAC` it is already hard to find the LCS by hand. Its length is 8, for example `CAGTGCAC`.

Checking all subsequences is not an option: a string of length k has up to 2^k subsequences, one for each subset of its positions, so brute force needs about $2^{\min\{|S|,|T|\}}$ checks.

7.4 Edit distance

How do we compare two strings? Option 1: transform one string into the other. Option 2: look at their subsequences. The two options turn out to be closely related.

Definition 7.4: Edit distance

Given two sequences S and T, the *edit distance* $\mathrm{ED}(S, T)$ is the minimum number of insertions and deletions of characters needed to transform S into T.

Example 25: Transforming a string

Let $S =$ `CATGA` and $T =$ `ACTGC`:

$$\texttt{CATGA} \xrightarrow{\text{delete C}} \texttt{ATGA} \xrightarrow{\text{insert C}} \texttt{ACTGA} \xrightarrow{\text{insert C}} \texttt{ACTGAC} \xrightarrow{\text{delete A}} \texttt{ACTGC}.$$

Four changes suffice, and in fact $\mathrm{ED}(S, T) = 4$.

We could also delete all the letters one by one and insert the new ones, but that is not the shortest way. When the strings are longer, it is hard to see the minimum number of steps.

The edit distance is symmetric, $\mathrm{ED}(S,T) = \mathrm{ED}(T,S)$, because we can perform all the steps in reverse: each insertion becomes a deletion and vice versa.

Note: Substitutions

Some definitions also allow *substitutions*, replacing one character with another at cost 1. With insertions and deletions only, a substitution costs 2: one deletion and one insertion.

7.4.1 Edit distance and LCS

A natural strategy for transforming S into T: find an LCS, delete everything else from S, leaving only the LCS, and then insert the characters of T that are missing.

Proposition 7.1: Edit distance from the LCS

$$\mathrm{ED}(S,T) = \underbrace{\big(|S| - \mathrm{LCS}(S,T)\big)}_{\text{characters to delete}} + \underbrace{\big(|T| - \mathrm{LCS}(S,T)\big)}_{\text{characters to insert}} = |S| + |T| - 2\,\mathrm{LCS}(S,T).$$

Why it holds. The strategy above uses exactly $|S| + |T| - 2\,\mathrm{LCS}(S,T)$ operations, so $\mathrm{ED}(S,T)$ is at most this number. Conversely, in any sequence of insertions and deletions, the characters of S that are never deleted appear in T in the same order: they form a common subsequence, of some length $k \le \mathrm{LCS}(S,T)$. We must then delete at least $|S| - k$ characters and insert at least $|T| - k$, so at least $|S| + |T| - 2k \ge |S| + |T| - 2\,\mathrm{LCS}(S,T)$ operations are needed. □

For Example 25, $\mathrm{LCS}(\texttt{CATGA}, \texttt{ACTGC}) = 3$ (for example `CTG`), so $\mathrm{ED} = 5 + 5 - 2 \cdot 3 = 4$.

7.5 A recursive formula for the edit distance

The optimal algorithm for computing the edit distance is not obvious. We now write a recursive formula, the starting point of *dynamic programming*.

Note: Notation

Consider two strings S and T with $|S| = n$ and $|T| = m$:

$$S = [S[1], S[2], \ldots, S[n]], \qquad T = [T[1], \ldots, T[m]].$$

We write $S^i = [S[1], \ldots, S[i]]$ and $T^j = [T[1], \ldots, T[j]]$ for the *prefixes* of length i and j (the superscript is not a power).

Proposition 7.2: Recurrence for the edit distance

$$\mathrm{ED}(S^i, T^j) = \begin{cases} j, & \text{if } i = 0, \\ i, & \text{if } j = 0, \\ \mathrm{ED}(S^{i-1}, T^{j-1}), & \text{if } S[i] = T[j], \\ \min\{\mathrm{ED}(S^i, T^{j-1}),\ \mathrm{ED}(S^{i-1}, T^j)\} + 1, & \text{if } S[i] \neq T[j]. \end{cases}$$

- If one string is empty, we must insert (or delete) all the characters of the other.
- If the last characters are equal, we do not need to modify them: we can ignore them and consider the distance between S^{i-1} and T^{j-1}. For example,

$$\mathrm{ED}(\texttt{CATGCAT T}, \texttt{ACCTAGAC T}) = \mathrm{ED}(\texttt{CATGCAT}, \texttt{ACCTAGAC}).$$

- If the last characters differ, one of them must go: either we remove the last character of S, giving $\mathrm{ED}(S^{i-1}, T^j) + 1$, or we remove the last character of T (equivalently, we insert it at the end of S), giving $\mathrm{ED}(S^i, T^{j-1}) + 1$. For `CATGCAT T` and `ACCTAGAC A`, we remove either the `T` or the `A`. We take the better of the two options.

Example 26: Small edit distances

- $\mathrm{ED}(\texttt{TGA}, \texttt{TTA}) = 2$: we must remove `G` and add `T`. Also $\mathrm{ED}(\texttt{TG}, \texttt{TT}) = 2$, because we have to remove a character and then add one.
- For `CAG` and `TGA`, the last characters differ. Either we remove `G` from `CAG`, which leaves `CA` and `TGA` (+1); or we remove `A` from `TGA`, which leaves `CAG` and `TG` (+1), whose last characters are now equal. Both options lead to $\mathrm{ED}(\texttt{CAG}, \texttt{TGA}) = 4$.

Algorithm 7.1: Recursive edit distance

```
1:  function ED(S^i, T^j)
2:      if i = 0 then
3:          return j
4:      else if j = 0 then
5:          return i
6:      else if S[i] = T[j] then
7:          return ED(S^{i-1}, T^{j-1})
8:      else
9:          return 1 + min{ED(S^i, T^{j-1}), ED(S^{i-1}, T^j)}
10:     end if
11: end function
```

Written directly as a recursive algorithm, this formula has an exponential running time.

Note: The same problem as Fibonacci

As with the recursive Fibonacci algorithm (Proposition 3.1), the recursion solves the same subproblems again and again. There are only $(n+1)(m+1)$ different subproblems $\mathrm{ED}(S^i, T^j)$, so storing each value once (dynamic programming) avoids the repeated work.